How we stop bleeding money on COD returns

To: Founder · From: Associate Software Engineer · Re: RTO decision engine · 30 Oct 2026

You asked for a system that looks at every COD order before it ships and decides what to do. I built it, and backtested it on the 2,017 real past COD orders you shared, which span **~6 months (1 Apr – 30 Sep 2026)**. The short version: **we should almost never cancel orders — that instinct actually loses money. The real wins come from confirming the medium-risk orders on WhatsApp and nudging the high-risk ones to prepaid.** Here is what the data says and what I recommend, in rupees.

> **Bottom line:** Over the ~6-month, 2,017-order sample the policy nets **+₹29,191** versus shipping everything blindly — that is **≈ ₹4,900 / month** at this sample's volume, or **≈ ₹14,500 per 1,000 COD orders**, so it scales with volume — while cancelling only **4 orders** (every one of which truly did RTO). Almost no genuine sales are lost.

1. Three things in the data that surprised me

Finding 1 **Cancelling "risky" orders is a money-losing idea for almost all of them.** A return costs us ₹169. But a delivered order earns ₹402 profit. The maths says a COD order is still worth shipping until its RTO chance passes **~70%**. Even an order with a coin-flip 50% RTO chance still makes **+₹116 on average**. So "cancel anything risky" would throw away profit on thousands of orders to dodge a few returns. Only a tiny sliver of orders are genuinely not worth shipping.

Finding 2 **Past behaviour predicts returns far better than anything else.** A customer who has returned **2+ orders before** returns again **76%** of the time — versus 27% for someone with a clean record. This one field is our strongest signal, and it is simple maths, not AI.

Finding 3 **Geography and address quality are huge — and address quality is the one an AI can read.** RTO ranges from **53% in Bihar** and 45% in UP down to **20% in Tamil Nadu**. And when the address has **no house/flat number** (just "Hanuman Mandir ke saamne"), RTO jumps to **40%** versus 23% for a complete address. That address judgment — especially messy Hinglish ones — is exactly where a language model earns its place.

2. The rules I recommend (and what they'd have done)

Every order gets a transparent 0–100 risk score from factors like return history, region, and address quality. The score maps to one of four actions. Applied to the full **~6-month sample (2,017 orders)**:

Action	Orders	Actual RTO in bucket	What we do
Ship as-is	1,043 (52%)	17%	Low risk — ship normally
Confirm on WhatsApp	739 (37%)	34%	Ask the customer to confirm first
Nudge to prepaid	231 (11%)	58%	Offer ₹40 off to pay online
Hold / cancel	4 (0.2%)	100%	Don't ship — certain loss

The buckets cleanly separate risk (17% → 34% → 58% → 100%), which is the proof the score works. Thresholds live in a plain config file your ops team can edit without a developer.

3. How I balanced Operations vs. Marketing (in ₹)

Operations wanted to auto-cancel risky COD; Marketing warned that cancelling a paid-ads first order wastes the ~₹300 we spent to acquire that customer. **The data sides mostly with Marketing** — cancelling is rarely worth it. So the system cancels only the ~0.2% of orders that are near-certain returns, and for everything risky in between it uses two softer, cheaper tools (figures below are over the full ~6-month sample):

- **WhatsApp confirmation** catches would-be returns without cancelling — it saved ~**₹21,600**, the single biggest lever.
- **Prepaid nudge** (₹40 off) converts high-risk COD to near-zero-risk prepaid — net ~**₹6,900** after discounts.
- A **guardrail** explicitly protects paid-ads first orders from auto-cancel unless the score is extreme (85+), directly honouring Marketing's CAC concern.

Why ₹40 discount: converting one risky COD order to prepaid saves ~₹137 in expected return loss, so ₹40 is comfortably profitable while still being attractive to the customer. The number lives in config — raise it for stubborn segments.

4. What could go wrong, and how we'd notice

- **The model drifts** (a new region or courier goes bad). *Notice:* watch actual RTO inside the "ship as-is" bucket — if it climbs past ~20%, the rules need retuning.
- **We over-confirm and annoy good customers.** *Notice:* track confirmation reply rate and opt-outs; if good customers complain, tighten the confirm threshold.
- **Address AI misreads a valid Hinglish address** and over-flags. *Notice:* sample 20 flagged addresses weekly; the engine logs its reason for every decision, so this is quick to audit.
- **Data labels are dirty** (the file had "Rto", "delivered", an "IN_TRANSIT"). I clean these, but garbage history would skew scores — worth a monthly data check.

5. What I'd build next with two more weeks

- Replace the hand-tuned score with a **trained model** (logistic regression / gradient boosting) on the same features — likely a few points better, still explainable.
- **Real WhatsApp integration** (the flow is already mocked and tested) plus a live prepaid payment link.
- **Auto-learning thresholds:** let the config update itself weekly from fresh outcomes, with a human approving the change.
- **Per-segment discounts** and an A/B test to find the true cheapest discount that still converts.

All figures are a backtest on the 2,017 COD orders provided (~6 months, 1 Apr – 30 Sep 2026), using the rules in config/rules.yaml. The net impact is reported three ways that reconcile — over the sample, per month (sample net ÷ ~6 months), and per 1,000 orders — so no single-month figure is ever implied from six months of data. Key assumptions: WhatsApp confirmation prevents 50% of would-be RTOs; 50% of nudged customers accept prepaid. These are deliberately conservative and easy to update as real data comes in.